# RPractical_Arellano,R

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A1.Create v as the sequence 5,10,15, . . . , 50 using seq(). Print its length, sum, and mean.

```
v <- seq(5, to = 50, by = 5)
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

A2. Create v that repeats the vector c(1,2,3) four times, and z that repeats each element of c(1,2,3) four times. Print both.

```
v <- rep(c(1,2,3), times = 4)
z <- rep(c(1,2,3), each = 4)
print(v)
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
print(z)
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3. Compute and print: the remainder of 17 divided by 5, the integer division of 17 by 5, 2 to the power of 10, square root of 144 and pi rounded to 3 decimals.

```
print(17 %% 5)
```

```
## [1] 2
```

```r
print(17 %/% 5)
```

```
## [1] 3
```

```r
print(2^10)
```

```
## [1] 1024
```

```r
print(sqrt(144))
```

```
## [1] 12
```

```r
x <- round(pi,3)
print(x)
```

```
## [1] 3.142
```

A4. Given a <- c(2,4,6,8) and b<- c(1,3,5,7), print a + b, a$b$, *the logical comparison* $a > b$, *abd the dot product sum (*a$b$)

```r
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)
a + b
```

```
## [1]  3  7 11 15
```

```r
a * b
```

```
## [1]  2 12 30 56
```

```r
a > b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a * b)
```

```
## [1] 100
```

A5. Use paste() to produce the "Student-1" "Student-2" "Student-3". Then identify the datatype of 42, 43L, "42", and TRUE. What is the output?

```r
paste("Student", 1:3, sep = "-")
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```r
class(42)
```

```
## [1] "numeric"
```

```r
class(42L)
```

```
## [1] "integer"
```

```r
class("42")
```

```
## [1] "character"
```

```r
class(41i)
```

```
## [1] "complex"
```

```r
class(TRUE)
```

```
## [1] "logical"
```

PART B: DATA FRAMES AND CONDITIONAL STATEMENTS

B1. Create a data frame students with 8 rows and these columns (use stringAsFactors = FALSE; sex is "M" or "F"):

```r
students <- data.frame(
id = 1:8,
name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
section = c("A", "B", "A", "C", "B", "C", "A", "B"),
midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
final = c(82, 79, 96, 71, 80, 91, 93, 65),
attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
stringsAsFactors = FALSE
)
```

B2. Display the structure, dimensions, column, names, and summary o students.

```r
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```r
dim(students)
```

```
## [1] 8 7
```

```r
colnames(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```r
summary(students)
```

```
##        id              name               sex         section     midterm
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                           Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3. Add a column average = the mean of the midterm and final for each student, rounded to 1 decimal.

```r
students$average <- round((students$midterm + students$final) / 2, 1)

print(students)
```

```
##   id  name sex section midterm final attendance average
## 1  1 Alice   F       A      78    82         95    80.0
## 2  2 Bruno   M       B      85    79         88    82.0
## 3  3  Chen   M       A      92    96        100    94.0
## 4  4  Dana   F       C      67    71         72    69.0
## 5  5  Elif   F       B      74    80         90    77.0
## 6  6 Farid   M       C      88    91         85    89.5
## 7  7 Grace   F       A      95    93         98    94.0
## 8  8  Hiro   M       B      59    65         60    62.0
```

B4. Add a logical column passed that is TRUE when the average >= 70 and attendance >= 75

```r
students$passed <- students$average >= 70 & students$attendance >= 75
print(students)
```

```
##   id  name sex section midterm final attendance average passed
## 1  1 Alice   F       A      78    82         95    80.0   TRUE
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE
```

```
## 3  3  Chen   M       A      92    96        100    94.0   TRUE
## 4  4  Dana   F       C      67    71         72    69.0  FALSE
## 5  5  Elif   F       B      74    80         90    77.0   TRUE
## 6  6 Farid   M       C      88    91         85    89.5   TRUE
## 7  7 Grace   F       A      95    93         98    94.0   TRUE
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE
```

B5. Using ifelse(), add column standing with these values based on average:

*"Honors" if average >= 90* "Good" ig average >= 75 (and below 90) *"Needs support" otherwise

```
students$standing <- ifelse(students$average >= 90, "Honors",
                     ifelse(students$average >= 75, "Good", "Needs                    Support"))
print(students)
```

```
##   id  name sex section midterm final attendance average passed
## 1  1 Alice   F       A      78    82         95    80.0   TRUE
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE
## 3  3  Chen   M       A      92    96        100    94.0   TRUE
## 4  4  Dana   F       C      67    71         72    69.0  FALSE
## 5  5  Elif   F       B      74    80         90    77.0   TRUE
## 6  6 Farid   M       C      88    91         85    89.5   TRUE
## 7  7 Grace   F       A      95    93         98    94.0   TRUE
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE
##                            standing
## 1                              Good
## 2                              Good
## 3                            Honors
## 4 Needs                     Support
## 5                              Good
## 6                              Good
## 7                            Honors
## 8 Needs                     Support
```

B6. Write a function that classify(score) that uses an if/else if/ else chain and returns "Distinction" for 85 above, "Merit" for 70 to 84, "Pass" for 50 to 69, and "fail" below 50. Test it with sapply(c(90,73, 55, 40)classify) and prnt results.

```
classify <- function(score) {
  if (score >= 85) {
    return("Distinction")
  } else if (score >= 70) {
    return("Merit")
  } else if (score >= 50) {
    return("Pass")
  } else {
    return("Fail")
  }
}
sapply(c(90, 73, 55, 40), classify)
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

Part C: rbind() and cbind()

C1. Create a data frame new_students with two students. Its columns must match the orginal, including average, passed and standing, computted the same way:

*id 9, Ines, F, section C, midterm 81, fina 84, attendance 92* id 10, Jamal,M, section A, midterm 70, final 68, attendance 80

Use rbind() to append them to students, saving the resuls as students_all

```
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE
)

new_students$average <- round((new_students$midterm + new_students$final) / 2, 1)
new_students$passed <- new_students$average >= 70 & new_students$attendance >= 75
new_students$standing <- ifelse(new_students$average >= 90, "Honors",
                         ifelse(new_students$average >= 75, "Good", "Needs Support"))

students_all <- rbind(students, new_students)
print(students_all)
```

```
##    id  name sex section midterm final attendance average passed
## 1   1 Alice   F       A      78    82         95    80.0   TRUE
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE
## 3   3  Chen   M       A      92    96        100    94.0   TRUE
## 4   4  Dana   F       C      67    71         72    69.0  FALSE
## 5   5  Elif   F       B      74    80         90    77.0   TRUE
## 6   6 Farid   M       C      88    91         85    89.5   TRUE
## 7   7 Grace   F       A      95    93         98    94.0   TRUE
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE
## 9   9  Ines   F       C      81    84         92    82.5   TRUE
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE
##                             standing
## 1                               Good
## 2                               Good
## 3                             Honors
## 4  Needs                     Support
## 5                               Good
## 6                               Good
## 7                             Honors
## 8  Needs                     Support
## 9                               Good
## 10                     Needs Support
```

C2. Create the numeric vector quiz <- c(8,7,10,6,7,9,10,5,8,6). Use cbind() to attach it to students_all as column names quiz. COnfirm the result is still a data frame with class().

```r
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz)
class(students_all)
```

```
## [1] "data.frame"
```

C3. Create the matrix m <- cbind(midterm = students-all$midterms, final = students$final) and use rbind() to append a final row containing the column means, labbeled "mean"

```r
m <- cbind(midterm = students_all$midterm, final = students_all$final)
m <- rbind(m, mean = colMeans(m))
m
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  68.0
## mean    78.9  80.9
```

C4. in a comment (2-3 sentences), explain what error rbind() produces if the data frames have different column names, and show the output if you safely catch it with tryCatch()

```r
bad <- data.frame(wrong_name = 1:2)
tryCatch(rbind(students, bad), error = function(e) {
  message("rbind failed: ", conditionMessage(e))
  NULL
})
```

```
## rbind failed: numbers of columns of arguments do not match
```

```
## NULL
```

PART D: SUBSETTING

Uses students_all(with quiz) for all questions. Print every results.

D1. Extract the name column in three ways: $, [["name]], and [, "name"].

```r
students_all$name
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
students_all[["name"]]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
students_all[, "name"]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

D2. Extract rows 2 to 5 and only the columns name, midterm, final.

```r
students_all[2:5, c("name", "midterm", "final")]
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

D3. Using logical subsetting, select students with attendance above 90.

```r
students_all[students_all$attendance > 90, ]
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 1  1 Alice   F       A      78    82         95    80.0   TRUE     Good    8
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 9  9  Ines   F       C      81    84         92    82.5   TRUE     Good    8
```

D4. Using subset(), select students of sex “F” in section “A” or “C” whose final is at least 75; return only name, section, and final

```r
subset(students_all, sex == "F" & section %in% c("A", "C") & final >= 75, select = c(name, section, fina
```

```
##    name section final
## 1 Alice       A    82
## 7 Grace       A    93
## 9  Ines       C    84
```

D5. Use which.max() and which.min() to report the name of the student wth the highest and lowest average.

```r
students_all$name[which.max(students_all$average)]
```

```
## [1] "Chen"
```

```r
students_all$name[which.min(students_all$average)]
```

```
## [1] "Hiro"
```

D6. Sort students_all by average in describing order, then print the top 3 rows.

```r
students_sorted <- students_all[order(students_all$average, decreasing = TRUE), ]
head(students_sorted, 3)
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 6  6 Farid   M       C      88    91         85    89.5   TRUE     Good    9
```

Part E: Basic Factors

E1. Convert section and sex in students-all to factors with factor(). Print levels() of each.

```r
students_all$section <- factor(students_all$section)
students_all$sex <- factor(students_all$sex)
levels(students_all$section)
```

```
## [1] "A" "B" "C"
```

```r
levels(students_all$sex)
```

```
## [1] "F" "M"
```

E2. Use table() to count students per sex and per section

```r
table(students_all$sex)
```

```
##
## F M
## 5 5
```

```r
table(students_all$section)
```

```
##
## A B C
## 4 3 3
```

```r
table(students_all$sex, students_all$section)
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

E3. Print as.integer() of the sex factor. In comment, explain what the integer represents

```r
as.integer(students_all$sex)
```

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

E4. Create a new factor sex_label from sex using factor() with levels =- c("F", "M") and labels = c("Female", "Male"). Add it to students_all and print its levels().

```r
students_all$sex_label <- factor(students_all$sex, levels = c("F", "M"), labels = c("Female", "Male"))
levels(students_all$sex_label)
```

```
## [1] "Female" "Male"
```

E5.Print summary() of sex_label. Then, using logical subsetting and mean(), print the mean average for each level of sex_label

```r
summary(students_all$sex_label)
```

```
## Female   Male
##      5      5
```

```r
mean(students_all$average[students_all$sex_label == "Female"])
```

```
## [1] 80.5
```

```r
mean(students_all$average[students_all$sex_label == "Male"])
```

```
## [1] 79.3
```